

Question Bank

Unit I: Basics of Automation and Control (09 Hrs.)

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Syllabus: Brief introduction about industrial processes and their automation; elements of pneumatic, hydraulic and electrical control systems; valves and actuators; stepper motors; PID controllers and their tuning.

Part A 10 multiple-choice questions $10 \times 1 = 10$ marks

Part B 5 short-answer questions $5 \times 2 = 10$ marks

Part C 3 long-answer questions $3 \times 10 = 30$ marks

Part A: Multiple-Choice Questions

1 mark each

Q1. The standard analog current signal used by industrial transmitters is

- (a) 4–20 mA
- (b) 0–5 mA
- (c) 10–50 mA
- (d) 0–100 mA

Answer: (a) 4–20 mA. The 4 mA “live zero” lets a broken wire (0 mA) be detected.

Q2. A pneumatic control system uses which working medium?

- (a) Oil
- (b) Water
- (c) Compressed air
- (d) Electric current

Answer: (c) Compressed air.

Q3. In a closed-loop control system, the error signal is

- (a) $SP + PV$
- (b) $SP \div PV$
- (c) $PV \times SP$
- (d) $SP - PV$

Answer: (d) $SP - PV$ (setpoint minus process variable).

Q4. A “5/2” directional control valve has

- (a) 5 positions and 2 ports
- (b) 5 ports and 2 positions
- (c) 5 valves and 2 actuators
- (d) 5 bar and 2 litres

Answer: (b) 5 ports and 2 switching positions.

Q5. Which valve protects a hydraulic system from excessive pressure?

- (a) Check valve
- (b) Needle valve
- (c) Relief valve
- (d) Ball valve

Answer: (c) Relief valve — it returns oil to the tank when the set pressure is exceeded.

Q6. FRL unit in a pneumatic system stands for

- (a) Filter, Regulator, Lubricator
- (b) Flow, Rate, Level
- (c) Force, Relief, Load
- (d) Fluid, Reservoir, Line

Answer: (a) Filter, Regulator, Lubricator.

Q7. A stepper motor with 200 steps per revolution has a step angle of

- (a) 0.9° (b) 7.5°
(c) 3.6° (d) 1.8°

Answer: (d) $360^\circ/200 = 1.8^\circ$.

Q8. Which term of a PID controller removes the steady-state error (offset)?

- (a) Proportional (b) Integral
(c) Derivative (d) None of these

Answer: (b) Integral — it keeps acting as long as any error remains.

Q9. Which actuator gives the largest force for its size?

- (a) Pneumatic cylinder (b) Solenoid
(c) Hydraulic cylinder (d) Stepper motor

Answer: (c) Hydraulic cylinder (working pressure 100–350 bar).

Q10. In the Ziegler–Nichols closed-loop tuning method, the controller is first set to

- (a) P-only (b) I-only
(c) D-only (d) Manual

Answer: (a) P-only; K_p is raised until sustained oscillation gives K_u and P_u .

Part B: Short-Answer Questions

2 marks each

Q1. Differentiate between open-loop and closed-loop control systems.

[2]

Answer

An **open-loop** system does not measure its output; the controller acts on the input alone (e.g. a washing-machine timer). A **closed-loop** (feedback) system measures the output, compares it with the setpoint and uses the error to correct the output (e.g. a thermostat). Closed loop is accurate and rejects disturbances; open loop is simpler and cheaper.

Q2. Why is 4 mA, not 0 mA, used as the lower limit of a 4–20 mA signal?

[2]

Answer

4 mA is a **live zero**: the zero of the measured range is represented by a non-zero current. If the wire breaks or the transmitter fails, the current falls to 0 mA, which is outside the valid range, so the fault is detected at once. The 4 mA also powers two-wire transmitters.

Q3. State Pascal's law. A force of 100 N acts on a piston of area 10 cm^2 ; find the force on a piston of area 200 cm^2 .

[2]

Answer

Pascal's law: pressure applied to a confined fluid is transmitted equally in all directions, so $F_1/A_1 = F_2/A_2$.

$$F_2 = F_1 \frac{A_2}{A_1} = 100 \times \frac{200}{10} = \mathbf{2000 \text{ N}}.$$

Q4. A stepper motor has a step angle of 1.8° and is driven at 1000 pulses/s. Find the steps per revolution and the speed in rpm.

[2]

Answer

$$\text{Steps/rev} = 360/1.8 = 200. \quad \text{Speed} = \frac{f \theta}{6} = \frac{1000 \times 1.8}{6} = 300 \text{ rpm (i.e. 5 rev/s).}$$

Q5. What is integral windup? How is it prevented?

[2]

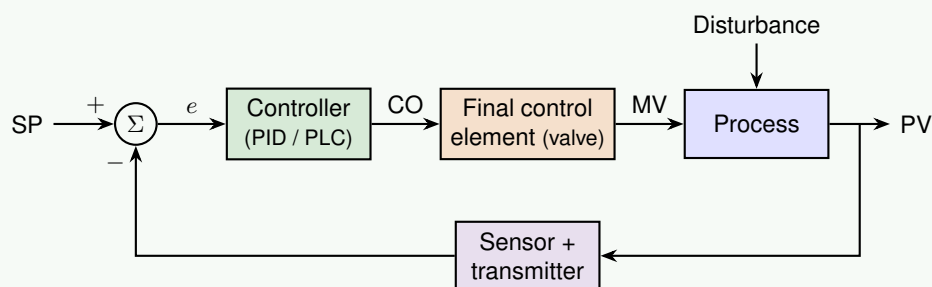
Answer

When the actuator saturates (e.g. a valve is already 100 % open) but the error persists, the integral term keeps growing. When the error reverses, the large stored integral takes a long time to “unwind”, causing big overshoot. It is prevented by **anti-windup**: clamping the integral, stopping integration while the output is saturated, or using the velocity form of PID.

Part C: Long-Answer Questions

10 marks each

Q1. With a neat block diagram, explain the elements of a closed-loop (feedback) control system. Compare it with an open-loop system and illustrate with the example of tank level control. **[10]**

Answer**(a) Block diagram****(b) Elements**

- **Setpoint (SP)** — the desired value of the controlled variable.
- **Summing junction / comparator** — forms the error $e = SP - PV$.
- **Controller** — decides the corrective output (CO) from the error (ON–OFF, PID, PLC logic).
- **Final control element** — valve, pump or motor that changes the manipulated variable (MV).
- **Process (plant)** — the equipment whose variable is controlled.
- **Sensor / transmitter** — measures the process variable (PV) and sends a standard signal (4–20 mA).
- **Disturbance** — any external change that upsets the PV (e.g. change in outflow).

(c) Open loop vs. closed loop

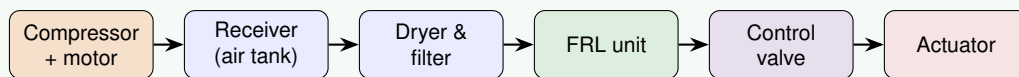
Feature	Open loop	Closed loop
Feedback	none	output measured and fed back
Accuracy	depends on calibration	high, self-correcting
Disturbances	not corrected	automatically corrected
Stability	always stable	can oscillate if badly tuned
Cost	low	higher (sensor + controller)
Example	toaster timer	thermostat, autopilot

(d) Example — tank level control: a level transmitter (LT) measures the water level and sends 4–20 mA to the level controller (LC). The LC compares it with the setpoint (say 1.5 m); if the level is low it opens the inlet control valve further, if high it closes it. If the outflow increases (disturbance), the level starts to fall, the error grows, and the controller opens the valve until the level returns to 1.5 m — without any operator action.

- Q2.** Explain the elements of a pneumatic system with a block diagram. Draw and explain the circuit of a double-acting cylinder controlled by a 5/2 single-solenoid valve. Calculate the extending and retracting forces of a cylinder of bore 50 mm and rod 20 mm at 6 bar. [10]

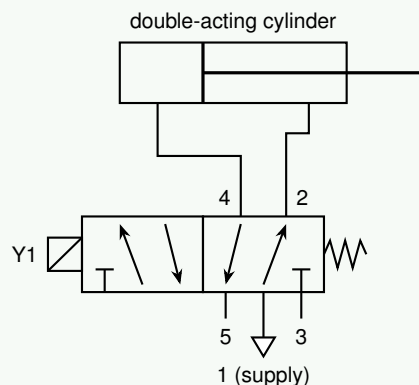
Answer

(a) Elements of a pneumatic system



- **Compressor** raises atmospheric air to 6–10 bar; the **receiver** stores air and damps pulsations.
- **Dryer and filter** remove moisture and dirt that would corrode and jam valves.
- **FRL**: filter (fine cleaning), regulator (constant working pressure), lubricator (oil mist for moving parts).
- **Directional control valves** route the air; flow control valves set the speed.
- **Actuators** (cylinders, air motors) convert air pressure into motion.

(b) Double-acting cylinder circuit



- **Rest position** (spring side): port 1 → 2 supplies the rod side, port 4 → 5 exhausts the cap side ⇒ the piston **retracts**.
- **Solenoid Y1 energised**: the valve shifts; 1 → 4 supplies the cap side, 2 → 3 exhausts ⇒ the piston **extends**.
- De-energising Y1 lets the spring return the valve, and the cylinder retracts.

(c) Force calculation ($p = 6 \text{ bar} = 0.6 \text{ N/mm}^2$)

$$A_{\text{cap}} = \frac{\pi}{4}(50)^2 = 1963 \text{ mm}^2 \Rightarrow F_{\text{ext}} = 0.6 \times 1963 \approx 1178 \text{ N}$$

$$A_{\text{ann}} = \frac{\pi}{4}(50^2 - 20^2) = 1649 \text{ mm}^2 \Rightarrow F_{\text{ret}} = 0.6 \times 1649 \approx 990 \text{ N}$$

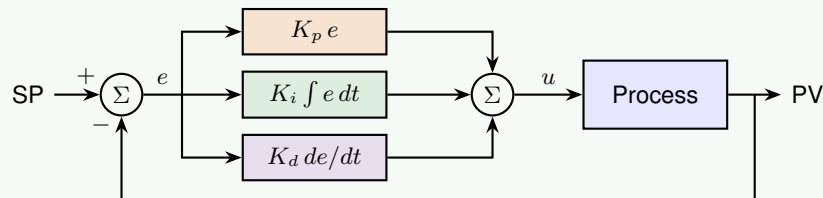
The retracting force is smaller because the rod occupies part of the piston area.

- Q3.** Draw the block diagram of a PID controller and explain the effect of the P, I and D actions. Describe the Ziegler–Nichols closed-loop tuning method with a flowchart, and find the PID settings for $K_u = 10$ and $P_u = 2$ s. [10]

Answer

(a) PID control law and block diagram

$$u(t) = K_p e(t) + K_i \int_0^t e d\tau + K_d \frac{de}{dt}, \quad K_i = \frac{K_p}{T_i}, \quad K_d = K_p T_d$$

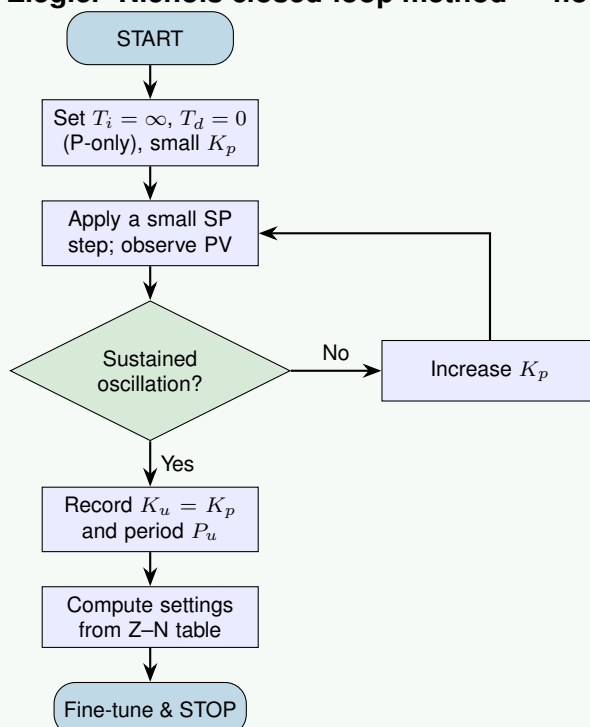


(b) Effect of each action

Increase	Rise time	Overshoot	Settling time	Steady-state error
K_p	decreases	increases	small change	decreases
K_i	decreases	increases	increases	eliminated
K_d	small change	decreases	decreases	no effect

P gives a fast response proportional to the present error but leaves an offset; **I** acts on the accumulated past error and removes the offset; **D** acts on the rate of change (predicts the future error), adds damping, but amplifies noise.

(c) Ziegler–Nichols closed-loop method — flowchart and settings



Controller	K_p	T_i	T_d
P	$0.5K_u$	∞	0
PI	$0.45K_u$	$P_u/1.2$	0
PID	$0.6K_u$	$P_u/2$	$P_u/8$

K_u = ultimate gain, P_u = period of the sustained oscillation.

(d) Numerical ($K_u = 10$, $P_u = 2$ s):

$$K_p = 0.6 \times 10 = 6, \quad T_i = 2/2 = 1 \text{ s}, \quad T_d = 2/8 = 0.25 \text{ s}$$

$$K_i = K_p/T_i = 6 \text{ s}^{-1}, \quad K_d = K_p T_d = 1.5 \text{ s}$$

These values give a fast response with about 25% overshoot and are used as a starting point for fine-tuning.